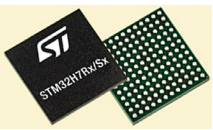
NEW PRODUCTS

TECHNOLOGY

COMPONENTS & MODULES

Microcontrollers with microprocessor functionality

STMicroelectronics has released the STM32H7 microcontroller units (MCUs), blending microprocessor functionality with microcontroller ease-of-use. Targeting sectors like smart appliances, building control, industrial automation, and personal medical devices, these MCUs facilitate advanced user interfaces and multitasking. They offer strong security features for IoT applications, including physical attack resistance and code isolation, aiming for top cyber security certifications.



STMicroelectronics
https://www.st.com/content

Wireless microcontrollers and SoCs

Silicon Labs has introduced the xG26 family, including the MG26, BG26, and PG26 devices, which feature advanced Cortex-M33 wireless SoCs and MCUs. This new series doubles the flash and RAM of its predecessor xG24, offering up to 3200kB and 512kB respectively, and includes more GPIO pins to support complex applications. They are equipped with a Matrix vector processor for AI/ML acceleration, catering to IoT, smart home, wearables, industrial automation, and automotive electronics. It targets non-wireless applications like CCTV and toys.

Silicon Labs
https://www.silabs.com

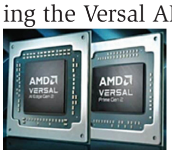
HMI microprocessor

Nuvoton has introduced the NuMicro MA35H0 series, a microprocessor unit designed for industrial human machine interface (HMI) applications. Ideal for sectors such as industrial HMI, factory automation, and smart buildings, this series integrates 128MB of DDR SDRAM into a compact 24mm[2] LQFP216 package, eliminating the need for external memory. This integration helps reduce space, electromagnetic interference, and costs. It includes a 2D graphics engine, video decoder, and supports various touchscreens, offering a secure, responsive, and cost-effective solution for HMI systems.

Nuvoton
https://www.nuvoton.com

Versal adaptive SoCs

AMD has unveiled the next generation of its Versal adaptive SoCs, introducing the Versal AI Edge Series Gen 2 and Versal Prime Series Gen 2. These devices are designed to enhance AI-driven embedded systems by integrating preprocessing, AI inference, and postprocessing in one chip. The new SoCs build on their predecessors with advanced AI engines that provide up to 3x more TOPs per watt and integrated ARM CPUs offering up to 10x the scalar compute power.



AMD
https://www.amd.com/en.html

Low-power IoT microcontrollers

Renesas Electronics Corporation has introduced the RA2A2 microcontroller group, featuring the ARM Cortex-M23 processor. These low-power MCUs are equipped with a 24-bit Sigma-Delta ADC and dual-bank code flash with bank swap functionality, enhancing firmware updates for IoT applications such as smart energy management, medical devices, and consumer electronics. The MCUs prioritise energy efficiency with multiple power modes and ultra-low power operation, boasting active mode energy consumption as low as 100µA/MHz and 0.40µA in software standby.

Renesas
https://www.renesas.com/us/en

Microcontrollers for better motor control

Toshiba Electronics Europe GmbH has broadened its TXZ+ Family Advanced Class with sixteen new 32-bit microcontrollers, featuring 1MB/512kB memory options across four package types, enhancing motor control for consumer and industrial applications. These units integrate an ARM Cortex-M4 processor capable of speeds up to 160MHz, coupled with durable flash memory that supports up to 100,000 cycles. They are designed for use in home appliances, HVAC systems, and industrial equipment, offering features like field-oriented control and the capability to drive up to three motors with a single MCU.

Toshiba
https://toshiba-india.com